SKY TECH ACADEMY

ROPE ACCESS TRAINING CENTER



ROPE ACCESS LEVEL 2

CANDIDATE TRAINING BINDER

SPRAT	IRATA
500 hrs + 6 months field	1,000 hrs + 12 months field
Upgrade from Level 1	Upgrade from Level 1

Houston, Texas | 2026

Section 1 — Level 2 Overview & Prerequisites

A Level 2 Rope Access Technician is required to know everything that a Level 1 knows, plus additional rigging and rescue techniques. Level 2 technicians must be able to work with limited supervision and therefore must have a real grasp on keeping others safe — through good rigging, good technique, good rescue ability, and a degree of managerial skill.

Each scheme requires that the Level 2 demonstrate an understanding of the responsibilities of a Level 2 Technician and how these fit into the overall responsibilities of the employer's Rope Access program.

1.1 Who This Binder Is For

- Moving from Level 1 to Level 2 — with minimum hours: 500 hrs + 6 months for SPRAT; 1,000 hrs + 12 months for IRATA
- Reviewing for renewal of a Level 2 Certificate
- Moving from Level 2 to Level 3 and refreshing previously learned skills

⚠ **Important:** *It is assumed that you are proficient in ALL Level 1 requirements before attending Level 2 training. If you are not proficient, delay your training and complete a refresher course first. Level 2 training builds directly on Level 1 foundations — gaps in Level 1 knowledge will significantly reduce your available training time for new Level 2 skills.*

1.2 Upgrade Requirements

Requirement	SPRAT	IRATA
Minimum Field Experience	6 months	12 months
Minimum Work Hours	500 logged hours	1,000 logged hours
Current Certification	Level 1 valid on assessment day	Level 1 valid on assessment day
Logbook	Must be signed and complete	Must be signed and complete
Training Required	Recommended (Affidavit required)	Mandatory — approved provider

Section 2 — Level 2 Assessment Checklist

The following checklist shows the maneuvers required for a Level 2 Rope Access Technician, above those already required for Level 1 certification. All Level 1 skills remain required and will not be re-listed here.

☐	Skill / Requirement	Scheme
☐	Knots and Hitches — friction hitches, Italian/Munter hitch, clove hitch, tensionless hitch	**SPRAT & IRATA**
☐	Load Sharing Anchors — wide Y-hang (anchors > 2 meters apart) with swing prevention	**SPRAT & IRATA**
☐	Vertical Aid Climbing — Upper, Lower, Center rhythm; two independent points of attachment at all times	**SPRAT & IRATA**
☐	Rigging for Hazard Avoidance — deviations, re-anchors, work restraint lines	**SPRAT & IRATA**
☐	Retrievable Rigging — pull-through anchor systems; rope protection; correct climbing side identification	**SPRAT & IRATA**
☐	Rope System Pre-Rigged to Lower — dual descender setup; lowering and belay mode operation	**SPRAT & IRATA**
☐	Cross Haul — two-system haul with rigging plate; team coordination and communication	**SPRAT & IRATA**
☐	Tensioned Lines — install at or above cross haul level; max tension one person on 3:1 system	**IRATA only**
☐	Rescue from Ascent Mode — pick-off from chest ascender; 4:1 mechanical advantage lift	**IRATA only**
☐	Rescue Past a Single Anchor Deviation	**IRATA only**
☐	Rescue Using a Rope to Rope Transfer with Casualty	**IRATA only**
☐	Rescue Through a Small Re-anchor with Casualty — re-anchor offset ≤ 1.5 meters	**IRATA only**
☐	Rescue From Fall Arrest Equipment	**IRATA only**
☐	Rescue Through Knots — casualty above knots; rescue then descend through knot system	**SPRAT only**
☐	Rescue From an Aid Climb — install ropes at casualty location; 5:1 MA lift; casualty into descent mode	**SPRAT & IRATA**
☐	Rescue Hauling — Pitch Head Haul system; rope grabs on working and safety lines	**SPRAT & IRATA**

Section 3 — Knots & Hitches

As a Level 2 technician, you are expected to know all Level 1 knots. This section covers the additional hitches required at Level 2. Friction hitches are generally a type of knot used to attach one rope to another in a way that is easily adjusted, but grips firmly when pulled.

3.1 Friction Hitches

Hitch	Description & Application
Prusik (Three-Wrap)	Friction hitch tied onto a rope; grips when loaded, slides when unloaded. Used as: rope grab in mechanical advantage systems; substitute for a rope grab device; emergency foot loop when ascending. NOTE: Do not use as the sole back-up in a system — strength varies with many variables.
Italian (Munter) Hitch	Adjustable friction hitch that functions as a belay or lowering device. Can substitute for a descender if yours fails. Must be tied off properly to prevent slippage. Also useful in rescue scenarios.
Clove Hitch	Adjustable hitch used in various rigging applications. DO NOT use on lifelines.
Tensionless Hitch	Maintains full rope strength when wrapped around a round structural member a minimum of 4 times. Begin with a barrel knot at the rope end, wrap 4+ times around the round member, clip carabiner through the barrel knot and around the rope.

Knots & Hitches — Level Requirements	Level 1	Level 2	Level 3	Level 1	Level 2	Level 3
Skill / Competency	**SPRAT**			**IRATA**		
Prusik (Three-wrap friction hitch) — tying, dressing, setting, and application		✓	✓		✓	✓
Italian (Munter) Hitch — tying, dressing, setting, and application as belay/descent device		✓	✓		✓	✓
Clove Hitch — tying and appropriate use		✓	✓		✓	✓
Tensionless Hitch — tying correctly around structural members with minimum 4 wraps		✓	✓		✓	✓

Section 4 — Advanced Rigging

Properly rigging ropes for different situations is critical to safety as a Level 2 technician. Always take time to consider all of the hazards you are trying to avoid and any new hazards your rigging may create.

4.1 Rigging Principles — Always Consider

1. Always rig systems that have at least two independent anchors
2. Try to share the load among your anchors — use a Double Figure 8 on a Bight for effective load distribution
3. Consider the entire rope path when evaluating hazards — not just the anchor point
4. Consider the swing potential when rigging offset anchor systems

4.2 Re-Anchors

Re-anchors are used to get around an obstacle or to avoid a hazard. Before installing a re-anchor, consider the following:

- Is there sufficient loop length? Insufficient loop makes passing difficult and rescue far more complex
- Have you actually avoided the hazard? Consider whether rope may contact the structure during transfer — particularly under platforms
- Do you have ropes leading to the ground on both sides? Install them where convenient — they are invaluable during rescue

4.3 Deviations

Deviations are used to avoid hazards or reposition ropes. Plan each deviation with the following questions:

- How great is the angle? Keep the angle at or below 20°. If greater than 20°, rig a re-anchor instead
- What is the potential for catastrophic failure? If a deviation failure could cause serious harm, rig a double deviation
- How ergonomic is the deviation for the technician to pass? If difficult, use longer anchor slings

4.4 Work Restraint Lines (IRATA)

Work restraint lines position a worker near a fall hazard without exposing them to it. Key rigging requirements:

- Rig lines overhead, as far from the fall hazard as possible
- Ensure the worker's lanyard (cow's tail) is short enough to reach the work area but not long enough to expose the worker to a fall
- Only one worker may be attached to any single segment of the line at a time

- Provide specific written work instructions for use

Work Restraint Lines	Level 1	Level 2	Level 3	Level 1	Level 2	Level 3
Skill / Competency	**SPRAT**			**IRATA**		
Rig and demonstrate work restraint lines in accordance with IRATA requirements					✓	✓

4.5 Load Sharing Anchors — Wide Y-Hang

For Y-hang systems where anchors are more than 2 meters apart, special precautions are required to prevent uncontrolled swing.

- Tie an alpine butterfly below the Y in each line and connect with a carabiner to prevent uncontrolled swing
- Extend the tail of the Figure 9 at least 3 feet (1 m) below the Y — these ropes are used to transfer to the Y-hang
- Keep all included angles well below 120° — 90° is recommended
- If using Double Figure 8s on a bight, the angle must not exceed 60°
- Always build redundancy into load-equalizing anchor systems to allow for failure of one leg

Load Sharing Anchors	Level 1	Level 2	Level 3	Level 1	Level 2	Level 3
Skill / Competency	**SPRAT**			**IRATA**		
Wide Y-hang system — anchors more than 2 meters apart; swing prevention; angle management		✓	✓		✓	✓
Demonstrate rigging and identifying correct angle loading for wide load sharing systems		✓	✓		✓	✓

4.6 Rope System Pre-Rigged to Lower

A rope system pre-rigged to lower allows a rescuer at the top anchor to safely lower a casualty. Goals of the system:

- Lower a casualty to a vertical hanging position with no obstructions below, or
- Lower the casualty next to a set of ropes without an obstruction below, or
- Lower the casualty to the ground for emergency medical attention, or
- Transport the casualty to a location where medical attention can be administered

Setup Procedure

1. Use two descenders — one as the lowering device, one in belay mode — prioritize safety over convenience
2. Ensure sufficient rope on the standing end to reach the final destination
3. Tie knots at the ends of both standing side ropes

4. Back-up each descender by tying a slip knot in the standing end approximately 1 meter from each device; connect the bight to the anchor

4.7 Retrievable Rigging — Pull-Through Anchor

A retrievable rigging system allows you to install an anchor and retrieve all ropes from the ground once work is complete.

Safe and Efficient Retrievable Rigging — Key Rules

5. Tie a knot in the pull side of the ropes to clearly indicate DO NOT CLIMB — ensure all technician entry points are also marked
6. Use appropriate rope protection on the climbing side of the system, secured to the pull side
7. Do not cross-load carabiners, especially across gates; particularly critical on non-rounded surfaces and small bore piping
8. Ensure you can reach both sides of the system from the ground before committing
9. Tie knots in the ends of all ropes, especially if they do not reach the ground

Sky Tech Academy: *Always climb on the correct side of a pull-through system — the side that will NOT release the anchors. The retrieval side must be clearly and visibly marked at every potential entry point.*

Advanced Rigging	Level 1	Level 2	Level 3	Level 1	Level 2	Level 3
Skill / Competency	**SPRAT**			**IRATA**		
Retrievable rigging — install, mark, and operate a pull-through anchor system		✓	✓		✓	✓
Rope system pre-rigged to lower — dual descender setup; lower and belay mode		✓	✓		✓	✓
Work restraint lines — rig correctly for fall prevention					✓	✓

Section 5 — Level 2 Rescue Operations

All Level 2 rescues build on the Level 1 rescue foundation. The key principle in all rescue through obstacles is to transfer from one rope system to another while the casualty is attached to you. Regardless of the obstacle — knots, deviations, or re-anchors — the underlying technique remains consistent.

5.1 Rescue Through Obstacles — Universal Rules

Critical Rules — Apply to ALL Rescue Through Obstacles:

- Ensure the short link connection between you and the casualty is positioned under all ropes — especially loops. Casualty and rescuer must never be on opposite sides of a loop
- Go shopping immediately after picking off the casualty — collect the casualty's descender, a back-up device, and spare carabiners before you need them
- Always use the casualty's cow's tail for the second connection to you — keep your own cow's tails free for fall factor management during transfers
- Do not tangle cow's tails — check the path of all cow's tails and ropes constantly. Find hold points and double-check every connection every time
- Always manage your fall factor — consult manufacturer's guidance for the back-up device in use. Keep back-up device as high as practical at all times

5.2 Rescue from Ascent Mode — Pick-Off from Chest Ascender (IRATA)

This rescue is required for IRATA Level 2. The challenge is lifting the casualty sufficiently to create slack in the chest ascender so it can be removed. A 4:1 mechanical advantage system is the recommended method at Sky Tech Academy.

Step-by-Step Procedure

10. Assess the casualty and confirm the rescue can be performed safely — do not proceed if you put yourself at risk
11. Connect your back-up safety device to the casualty's working rope and your ascenders to the casualty's safety rope — this allows equal loading of both ropes during ascent
12. Ascend to the casualty until level; place your spare back-up device above the casualty's ascender and push all the way up; push casualty's hand ascender up too — maintain 4 inch clearance above toothed devices
13. Remove your first back-up device; ascend past the casualty and change over into your descender; lock off the descender; clear ropes from between you and the casualty
14. Establish two points of contact: (1) clip casualty's cow's tail into your ventral attachment; (2) connect short link from your descender carabiner to casualty's sternal attachment — short link must point downward
15. Confirm two points of contact then remove casualty's hand ascender and back-up device

16. Attach a rope grab above casualty's chest ascender on the working line; add a pulley with carabiner; using accessory cord, run through casualty's central attachment, through the pulley, and back twice — creating a 4:1 mechanical advantage system
17. Pull cord to remove all slack between the harness and the pulley/rope grab; add hand ascender to cord below pulley; shorten foot-loop
18. Keep your back-up device as high as practical; reach down and grasp the casualty at the waist/ventral attachment; pull up with arms while pushing down with foot — lift casualty enough to remove chest ascender
19. Apply friction to the mechanical system by squeezing the cord at the bottom of the pulley; remove hand ascender while holding friction; lower casualty onto the short link; remove lifting system
20. Add an extra carabiner to your descender carabiner on the side facing you; feed the tail rope through this carabiner as additional friction; maintain fall factor below 0.5; descend slowly, keeping the casualty from fully lying on the ground — move legs to restore circulation

> ⚠ **Suspension Trauma:** *Never allow all weight to be removed from the casualty at once after prolonged suspension. Sudden pressure release can cause blood to surge through arteries causing additional injury. Lower slowly and move the casualty's legs continuously to restore circulation.*

5.3 Rope to Rope Transfer with Casualty (IRATA)

A relatively straightforward technique. The casualty is already attached to you after a pick-off. There will be no obstacle in the way during the transfer itself.

21. Pick off casualty; go shopping — collect casualty's descender
22. Add casualty's descender to your harness, capturing your ventral D-ring and the short link connection; position the descender on the side of your D-ring in the direction of travel
23. Place a safety back-up on the transfer ropes; pull all slack out of the safety line
24. Place the transfer rope into your new descender and remove all slack
25. Install a lifting system: place a rope grab above your descender; run the standing end through a pulley; attach the pulley to the rope grab — creates a variable ratio lift
26. Place an ascender with foot-loop on the pull rope and climb toward the far anchors
27. Once obstructions are cleared, descend using your original descender to the final destination

5.4 Rescue Through a Small Re-anchor with Casualty (IRATA)

A re-anchor is essentially a series of rope to rope transfers. The casualty will be on the far side of the re-anchor. Approach using the casualty's safety line as your working line.

28. Pick off the casualty onto your descender carabiner; go shopping; ensure ropes are not under the short link
29. Descend if above the far anchors — position your descender slightly lower than the far anchors
30. Add casualty's descender to your harness — capture ventral D-ring and short link; position in the direction of travel

31. Place a safety back-up on the transfer ropes; pull all slack from safety line without tightening the cow's tail
32. Place transfer rope into new descender and remove all slack
33. Install a lifting system; climb toward far anchors — do NOT descend into the bottom of the loop (this creates dangerous loading conditions)
34. Once near far anchors, use original descender to lower until load transfers to new descender; remove original descender and back-up
35. With exit ropes available, check everything is untangled and not under short link; reposition original descender to capture both central attachment and short link; add extra back-up to exit ropes
36. Complete another transfer and descend to the ground

5.5 Rescue Past Knots (SPRAT)

The casualty will be above the knots. Ascend past the casualty, perform a rescue (descent or ascent mode depending on equipment), go shopping, then descend toward the knots.

37. Ascend past casualty; perform rescue and establish two points of contact; go shopping (collect descender and back-ups)
38. Descend on working line until the knot below your descender is approximately 60 cm (24 inches) from your descender — allow extra distance for rope stretch on long drops
39. On the safety line, move the existing knot above the height of your descender — take up slack from above the knot and feed into the bight to isolate the damaged section
40. Add casualty's descender to harness — capture ventral D-ring and short link connection
41. Place a back-up on your working line above your descender (making twins)
42. Feed the rope below the knot in the safety line into the new descender; remove as much slack as possible
43. Transfer the load from the old descender to the new by operating the old descender (use additional friction for two-person load)
44. Once transferred, remove the slack descender; remove original back-up; descend down
45. Manage fall factor carefully with two-person load — keep back-up device high at all times
46. Negotiate the knot on your safety line by positioning back-up devices appropriately

5.6 Rescue Past Deviation (IRATA)

Similar in principle to rescue through knots — transferring from the working line to the safety line. This technique works for both single and double deviations.

47. With casualty attached, descend until the top of your descender is slightly lower than the lowest deviation anchor connection
48. Using the ropes above the deviation, pull yourself into position at the deviation; grab the ropes coming from below the deviation; connect them to your harness temporarily
49. Place a back-up above your descender on your working line (making twins)
50. Add casualty's descender to harness — capture ventral D-ring and short link; position on the side closest to the deviation

51. Install the safety rope (coming from below the deviation) into the new descender; pull all slack out — move toward the deviation, aiming for approximately 10 cm (4 inches) clearance
52. Lower from old descender to transfer load to safety line below deviation; take off old descender once slack
53. Manage both back-up devices as you descend — remove first back-up and descend slightly to position below deviation
54. Add a safety back-up below the deviation; remove the back-up above the deviation; descend to final destination

5.7 Rescue From Fall Arrest Equipment (IRATA)

Rescuing a casualty who has taken a fall in fall arrest equipment. The quickest approach is rescue from a separate rope system.

55. Aid climb to the location where the casualty is suspended
56. Install a rope system and attach your descent equipment
57. Descend to the casualty and establish two points of contact as in any other rescue
58. Rig a simple lifting system and lift the casualty, creating slack in their fall arrest lanyard
59. Unclip the fall arrest lanyard from the casualty's harness
60. Lower the casualty onto your short link connection
61. Descend with the casualty to a safe area

5.8 Rescue From an Aid Climb

Rescue from aid climb is particularly challenging because the rescuer is typically at the same level as or below the casualty, reducing mechanical advantage. A 5:1 mechanical advantage system is recommended.

62. Aid out to the casualty's location
63. Install a set of ropes at the casualty's location — one rope anchored to the casualty's anchor, one to your anchor
64. Immediately place the rope at the casualty's anchor into the casualty's descender and remove all slack; place a safety back-up from the casualty's harness on the rope next to you — the casualty is now connected to the newly installed ropes
65. Install a 5:1 mechanical advantage lifting system using accessory cord
66. Lift the casualty and release the chest ascender — or alternatively release the carabiner the cow's tail is connected to the anchor with (if releasing the carabiner, ensure it is on the front or back of the anchor, never trapped between two loaded carabiners)
67. The casualty is now in descent mode on the ropes you installed — this limits how far you need to lower them
68. Get onto the opposite ropes and perform a descent rescue
69. Perform a rope to rope transfer with the casualty to reach the final destination if required

5.9 Rescue Hauling — Pitch Head Haul

A pitch head haul system is useful for quickly raising a casualty to a platform, or with modification, to another accessible area. The rescuer operates from the platform above.

Key Differences from Level 1 Hauling System

70. Do not re-direct ropes through pulleys over the casualty — attach your lowering and lifting system directly over the casualty instead
71. Use rope grabs to attach your system to the casualty's ropes rather than tying directly to the load
72. The rope grab on the working line must be a toothed device (e.g. hand ascender) — position all the way down the working line to the casualty's working equipment
73. The rope grab on the safety line must be a non-aggressive device — position within reach but with clear separation from the safety back-up

⚠ **Load Management:** *Always pay attention to system loading. Do not make a bad situation worse by over-stressing a system beyond its Safe Working Load. In complex rescues, consider pre-rigging and testing the system before the access takes place.*

5.10 Cross Haul

A cross haul transports a casualty (or load) horizontally between two anchor systems. The Level 2 may be involved in or required to supervise a team cross haul. Clear roles, planning, and communication are essential.

Setup

74. Install two haul systems — one anchored over the load, one over the intended landing area
75. Attach the ends of each haul system to the load using a rigging plate — the rigging plate is essential for multi-directional loading
76. Ensure the rigging plate is backed up along with the rest of the system

Planning Considerations

- What is the included angle at the height of the haul? Greater angle = greater force on each side of the system
- What is the anticipated total load? Double up on descenders if load is heavy or angles produce elevated forces
- Are tensions kept equal throughout the system during lowering, especially when load sharing among anchors?
- How many people are needed to safely operate both systems?

Operation

- Operate one side of the system at a time
- Lift the load first, move to the intended landing area clearing obstacles, then lower using the other system
- Always keep the load as low as necessary to reduce the included angle

5.11 Tensioned Lines (IRATA)

Tensioned lines distribute forces among many anchors and define the load path. They work in conjunction with a cross haul system — they do not replace it.

- Install a cross haul first — cross haul systems are the motor behind the operation (lifting, transporting, and lowering)
- Install tensioned lines at the same level as or above the cross haul, following the same load path
- Rig tension lines as high as practicable
- Keep the load as low as possible — remember angle loading increases with height
- Do not over-tension lines: maximum tension is one person pulling on a 3:1 mechanical advantage system — never use more than two people pulling on a 3:1 to tension lines
- Always install at least one control line on the bottom side of a guideline
- During transport, operate the cross haul correctly — use mechanical advantage on the haul side; put descender(s) into belay mode on the release side

5.12 Vertical Aid Climbing

Vertical aid climb is performed using the same rhythm as horizontal aid climb, adapted for vertical movement. The same principle applies: maintain two independent points of attachment at all times with fall factor < 1.

Horizontal Aid Climb Rhythm	Vertical Aid Climb Rhythm
Moving Right: Right → Left → Center → Repeat **Moving Left:** Left → Right → Center → Repeat	**Moving Up:** Upper → Lower → Center → Repeat *Key principle: two independent points of contact at all times*

Section 6 — Level 2 Skills Matrix

The following matrix shows all skills required at Level 2 for both SPRAT and IRATA. All Level 1 skills remain applicable. A filled cell with ✓ indicates the skill is required at that level for that scheme.

Knots & Hitches	Level 1	Level 2	Level 3	Level 1	Level 2	Level 3
Skill / Competency	**SPRAT**			**IRATA**		
Prusik (Three-wrap) friction hitch — tying, dressing, setting, applications		✓	✓		✓	✓
Italian (Munter) Hitch — belay and descent device applications		✓	✓		✓	✓
Clove Hitch — tying and appropriate rigging applications		✓	✓		✓	✓
Tensionless Hitch — minimum 4 wraps around a round member		✓	✓		✓	✓

Advanced Rigging	Level 1	Level 2	Level 3	Level 1	Level 2	Level 3
Skill / Competency	**SPRAT**			**IRATA**		
Wide load sharing anchor (Y-hang) — anchors > 2 m apart; swing prevention; angle management		✓	✓		✓	✓
Rigging for hazard avoidance — deviations and re-anchors		✓	✓		✓	✓
Retrievable rigging — pull-through anchor system; marking; rope protection		✓	✓		✓	✓
Rope system pre-rigged to lower — dual descender; lowering and belay mode		✓	✓		✓	✓
Work restraint lines — rig for fall prevention with appropriate lanyard					✓	✓
Vertical aid climbing — Upper / Lower / Center rhythm; FF < 1; two independent attachments		✓	✓		✓	✓

Cross Haul & Tensioned Lines	Level 1	Level 2	Level 3	Level 1	Level 2	Level 3
Skill / Competency	**SPRAT**			**IRATA**		
Cross haul — two haul systems with rigging plate; team coordination		✓	✓		✓	✓
Tensioned lines — install at or above cross haul level; max tension 1 person on 3:1					✓	✓

Level 2 Rescue Operations	Level 1	Level 2	Level 3	Level 1	Level 2	Level 3
Skill / Competency	**SPRAT**			**IRATA**		
Pick-off from chest ascender (ascent mode) — 4:1 MA lift to release chest ascender					✓	✓
Rescue using rope to rope transfer with casualty					✓	✓
Rescue through a small re-anchor with casualty — offset ≤ 1.5 m					✓	✓
Rescue past a single anchor deviation with casualty					✓	✓
Rescue from fall arrest equipment					✓	✓
Rescue through knots — casualty above knots; descent rescue then negotiate knots		✓	✓			
Rescue from an aid climb — install ropes at casualty location; 5:1 MA lift; descent rescue		✓	✓		✓	✓
Rescue hauling — Pitch Head Haul system; rope grabs on working and safety lines		✓	✓		✓	✓

This training binder is proprietary and confidential. All rights reserved — Sky Tech Academy 2026.